```sql
WITH flights_with_airports AS (
    SELECT
        f.flight_id,
        da.country ->> 'ru' AS dep_country,
        aa.country ->> 'ru' AS arr_country
    FROM flights f
    JOIN routes r
        ON r.route_no = f.route_no
    JOIN airports_data da
        ON da.airport_code = r.departure_airport
    JOIN airports_data aa
        ON aa.airport_code = r.arrival_airport
),
inter AS (
    SELECT *
    FROM flights_with_airports
    WHERE dep_country <> arr_country
),
counts AS (
    SELECT
        dep_country,
        arr_country,
        COUNT(*) AS flights_cnt
    FROM inter
    GROUP BY dep_country, arr_country
),
pairs AS (
    SELECT
        LEAST(dep_country, arr_country)    AS country_a,
        GREATEST(dep_country, arr_country) AS country_b,
        dep_country,
        arr_country,
        flights_cnt
    FROM counts
),
pair_totals AS (
    SELECT
        country_a,
        country_b,
        SUM(flights_cnt) AS total_between
    FROM pairs
    GROUP BY country_a, country_b
)
SELECT
    dep_country || ' -> ' || arr_country AS route_countries,
    flights_cnt
FROM pairs p
JOIN pair_totals t
  ON t.country_a = p.country_a
 AND t.country_b = p.country_b
ORDER BY
    t.total_between DESC,
    t.country_a,
    t.country_b,
    flights_cnt DESC;
```

Рисунок 1. Код. Задание 1

	route_countries text	flights_cnt bigint
1	Соединенное Королевство -> Германия	435
2	Германия -> Соединенное Королевство	435
3	Франция -> Германия	431
4	Германия -> Франция	430
5	Франция -> Соединенное Королевство	335
6	Соединенное Королевство -> Франция	335
7	Италия -> Германия	246
8	Германия -> Италия	244
9	Китай -> Непал	220
10	Непал -> Китай	218
11	Швеция -> Соединенное Королевство	210
12	Соединенное Королевство -> Швеция	210
13	Соединенное Королевство -> Италия	184
14	Италия -> Соединенное Королевство	182
15	Китай -> Япония	160
16	Япония -> Китай	158
17	Франция -> Австрия	147
18	Австрия -> Франция	147
19	Соединенные Штаты -> Канада	128
20	Канада -> Соединенные Штаты	127
21	Франция -> Италия	97

Всего строк: 136 Запрос завершен 00:00:00.055

Рисунок 2. Результат. Задание 1

```sql
WITH ticket_seg AS (
    SELECT
        t.passenger_name,
        t.book_ref,
        t.ticket_no,
        f.route_no,
        f.scheduled_departure,
        COUNT(*) OVER (PARTITION BY t.ticket_no) AS seg_cnt
    FROM tickets t
    JOIN boarding_passes bp
        ON bp.ticket_no = t.ticket_no
    JOIN flights f
        ON f.flight_id = bp.flight_id
)
SELECT
    passenger_name,
    book_ref,
    ticket_no,
    route_no,
    ROW_NUMBER() OVER (
        PARTITION BY ticket_no
        ORDER BY scheduled_departure
    ) AS transfer_no
FROM ticket_seg
WHERE seg_cnt > 1
ORDER BY
    passenger_name,
    ticket_no,
    transfer_no;
```

Рисунок 3. Код. Задание 2

	passenger_name text	book_ref character (6)	ticket_no text	route_no text	transfer_no bigint
1	Aada Heikkinen	DDG3RR	00054324013...	PG0315	1
2	Aada Heikkinen	DDG3RR	00054324013...	PG0124	2
3	Aada Koivuniemi	C42SCM	00054331793...	PG0315	1
4	Aada Koivuniemi	C42SCM	00054331793...	PG0124	2
5	Aada Koivuniemi	C42SCM	00054331793...	PG0123	1
6	Aada Koivuniemi	C42SCM	00054331793...	PG0316	2
7	Aada Siitonen	LWHP9K	00054326133...	PG0315	1
8	Aada Siitonen	LWHP9K	00054326133...	PG0182	2
9	Aada Siitonen	LWHP9K	00054326133...	PG0358	3
10	Aanya Aggarwal	TQ5EOK	00054325849...	PG0373	1
11	Aanya Aggarwal	TQ5EOK	00054325849...	PG0051	2
12	Aanya Agrawal	FVTUON	00054329528...	PG0376	1
13	Aanya Agrawal	FVTUON	00054329528...	PG0373	2
14	Aanya Agrawal	FVTUON	00054329528...	PG0374	1
15	Aanya Agrawal	FVTUON	00054329528...	PG0375	2
16	Aanya Anand	NQ7O0X	00054324852...	PG0297	1
17	Aanya Anand	NQ7O0X	00054324852...	PG0112	2
18	Aanya Babar	XUQ2X3	00054330761...	PG0337	1
19	Aanya Babar	XUQ2X3	00054330761...	PG0188	2
20	Aanya Babu	ASBYLN	00054340198...	PG0051	1

Всего строк: 880802 Запрос завершен 00:00:03.807

Рисунок 4. Результат. Задание 2

```sql
WITH late_per_flight AS (
    SELECT
        f.flight_id,
        f.route_no,
        date_part('year', f.scheduled_departure)  AS year,
        date_part('month', f.scheduled_departure) AS month,
        ad.country ->> 'ru' AS dep_country,
        COUNT(*) FILTER (
            WHERE bp.boarding_time > f.scheduled_departure
        ) AS late_pas
    FROM flights f
    JOIN boarding_passes bp
        ON bp.flight_id = f.flight_id
    JOIN routes r
        ON r.route_no = f.route_no
    JOIN airports_data ad
        ON ad.airport_code = r.departure_airport
    GROUP BY
        f.flight_id,
        f.route_no,
        year,
        month,
        dep_country
),
agg AS (
    SELECT
        year,
        month,
        dep_country,
        COUNT(*) AS flights_total,
        AVG(late_pas::numeric) AS avg_late_pas,
        MAX(late_pas) AS max_late_pas
    FROM late_per_flight
    GROUP BY
        year,
        month,
        dep_country
)
SELECT
    a.year,
    a.month,
    a.dep_country AS departure_country,
    a.flights_total,
    a.avg_late_pas,
    a.max_late_pas,
    l.route_no AS route_no_max_late
FROM agg a
JOIN LATERAL (
    SELECT
        lpf.route_no
    FROM late_per_flight lpf
    WHERE
        lpf.year = a.year
        AND lpf.month = a.month
        AND lpf.dep_country = a.dep_country
    ORDER BY lpf.late_pas, lpf.route_no
    LIMIT 1
) AS l ON TRUE
ORDER BY
    a.year,
    a.month,
    a.dep_country;
```

Рисунок 5. Код. Задание 3

	year double precision	month double precision	departure_country text	flights_total bigint	avg_late_pas numeric	max_late_pas bigint	route_no_max_late text
1	2025	10	Австрия	49	0.00000000000000000000	0	PG0094
2	2025	10	Германия	301	40.6810631229235880	1616	PG0024
3	2025	10	Индия	853	25.6576787807737397	1616	PG0019
4	2025	10	Италия	252	40.0714285714285714	1616	PG0085
5	2025	10	Канада	406	56.5837438423645320	1616	PG0007
6	2025	10	Китай	1230	32.8325203252032520	1616	PG0028
7	2025	10	Непал	27	2.6666666666666667	39	PG0039
8	2025	10	Новая Зеландия	26	3.0000000000000000	78	PG0089
9	2025	10	Россия	622	14.6752411575562701	1300	PG0003
10	2025	10	Соединенное Королевст...	163	20.2515337423312883	975	PG0120
11	2025	10	Соединенные Штаты	917	28.6259541984732824	1212	PG0137
12	2025	10	Финляндия	57	5.5438596491228070	160	PG0070
13	2025	10	Франция	235	30.4000000000000000	1616	PG0069
14	2025	10	Чешская Республика	39	2.2564102564102564	88	PG0081
15	2025	10	Чили	66	0.00000000000000000000	0	PG0001
16	2025	10	Швеция	53	4.3207547169811321	229	PG0117
17	2025	10	Япония	246	53.2520325203252033	1616	PG0057
18	2025	11	Австрия	30	14.8666666666666667	404	PG0357
19	2025	11	Германия	321	33.6791277258566978	1616	PG0121
20	2025	11	Индия	877	27.2257696693272520	1212	PG0019
21	2025	11	Италия	227	39.9162995594713656	1616	PG0085
22	2025	11	Канада	398	56.6180904522613065	1616	PG0007

Всего строк: 51 | Запрос завершен 00:00:02.883

Рисунок 6. Результат. Задание 3

```sql
WITH passenger_flights AS (
    SELECT
        t.passenger_id, t.passenger_name, bp.ticket_no,
        f.flight_id, f.route_no, f.scheduled_departure,
        extract(year from f.scheduled_departure) AS year,
        r.departure_airport, r.arrival_airport,
        ad_dep.timezone AS dep_tz,
        ad_arr.timezone AS arr_tz,
        ad_arr.country ->> 'ru' AS arr_country_ru
    FROM task4_passengers p
    JOIN tickets t
        ON t.passenger_id = p.passenger_id
    JOIN boarding_passes bp
        ON bp.ticket_no = t.ticket_no
    JOIN flights f
        ON f.flight_id = bp.flight_id
    JOIN routes r
        ON r.route_no = f.route_no
    JOIN airports_data ad_dep
        ON ad_dep.airport_code = r.departure_airport
    JOIN airports_data ad_arr
        ON ad_arr.airport_code = r.arrival_airport
),
stats_base AS (
    SELECT
        passenger_id,
        MAX(passenger_name) AS passenger_name,
        year,
        COUNT(DISTINCT flight_id) AS flights_cnt,
        SUM( (dep_tz IS DISTINCT FROM arr_tz)::int ) AS tz_changes
    FROM passenger_flights
    GROUP BY passenger_id, year
),
final_countries_agg AS (
    SELECT
        passenger_id,
        year,
        COUNT(DISTINCT final_country) AS final_countries_cnt
    FROM final_countries
    GROUP BY passenger_id, year
)
SELECT
    sb.passenger_id, sb.passenger_name, sb.year,
    sb.flights_cnt  AS flights_total,
    COALESCE(fca.final_countries_cnt, 0) AS final_countries_total,
    sb.tz_changes AS timezone_changes_total
FROM stats_base sb
LEFT JOIN final_countries_agg fca
  ON fca.passenger_id = sb.passenger_id
 AND fca.year = sb.year
ORDER BY
    sb.passenger_id,
    sb.year;
```

Рисунок 7. Код. Задание 4

	passenger_id text	passenger_name text	year numeric	flights_total bigint	final_countries_total bigint	timezone_changes_tot bigint
1	AT 02081671700...	Kurt Hafner	2025	6	2	
2	AT 02745327414...	Eduard Steindl	2025	6	2	
3	AT 03833395918...	Maria Posch	2025	4	2	
4	AT 04284911363...	Miriam Schmidt	2025	2	2	
5	AT 04431381749...	Gernot Kellner	2025	5	2	
6	AT 07031405092...	Doris Eberharter	2025	6	2	
7	AT 07926996239...	Margarete Weber	2025	6	2	
8	AT 08500701640...	Greta Schöpf	2025	4	2	
9	AT 13796071907...	Thomas Weiß	2025	6	2	
10	AT 14191607015...	Albert Krenn	2025	4	2	
11	AT 14451521146...	Ursula Reiter	2025	4	2	
12	AT 15864672573...	Siegfried Auer	2025	8	2	
13	AT 18319541123...	Herta Lamprecht	2025	6	2	
14	AT 18915935277...	Christoph Rabl	2025	6	2	
15	AT 22388691307...	Heinrich Krainer	2025	6	2	
16	AT 22476482145...	Manuela Heindl	2025	7	2	
17	AT 25320731070...	Andreas Schön	2025	4	2	

Всего строк: 10000 Запрос завершен 00:00:00.386

Рисунок 8. Результат. Задание 4

```sql
WITH flights_per_plane AS (
    SELECT
        date_part('year', f.scheduled_departure)  AS year,
        date_part('month', f.scheduled_departure) AS month,
        r.airplane_code,
        COUNT(*) AS flights_cnt
    FROM flights f
    JOIN routes r
        ON r.route_no = f.route_no
    GROUP BY
        date_part('year', f.scheduled_departure),
        date_part('month', f.scheduled_departure),
        r.airplane_code
)
SELECT
    fpp.year, fpp.month,
    ad.model ->> 'ru' AS airplane_name_ru,
    fpp.flights_cnt,
    1 + (
        SELECT COUNT(*)
        FROM flights_per_plane fpp2
        WHERE
            fpp2.year  = fpp.year
            AND fpp2.month = fpp.month
            AND fpp2.flights_cnt > fpp.flights_cnt
    ) AS rank_in_month
FROM flights_per_plane fpp
JOIN airplanes_data ad
    ON ad.airplane_code = fpp.airplane_code
ORDER BY
    fpp.year, fpp.month,
    rank_in_month, airplane_name_ru;
```

Рисунок 9. Код. Задание 5

	year double precision	month double precision	airplane_name_ru text	flights_cnt bigint	rank_in_month bigint
1	2025	10	Боинг 777-300ER	4936	1
2	2025	10	Боинг 787-9 Dreamli...	2012	2
3	2025	10	Аэробус A350-1000	1870	3
4	2025	10	Боинг 737 MAX 7	1859	4
5	2025	10	Эмбраэр E170	1325	5
6	2025	10	Аэробус A330-900neo	1049	6
7	2025	10	Бомбардье CRJ700	602	7
8	2025	10	Аэробус A320neo	505	8
9	2025	11	Боинг 777-300ER	4739	1
10	2025	11	Боинг 737 MAX 7	1940	2
11	2025	11	Боинг 787-9 Dreamli...	1885	3
12	2025	11	Аэробус A330-900neo	1433	4
13	2025	11	Аэробус A350-1000	1411	5
14	2025	11	Эмбраэр E170	935	6
15	2025	11	Аэробус A320neo	626	7
16	2025	11	Бомбардье CRJ700	616	8
17	2025	12	Боинг 777-300ER	5099	1
18	2025	12	Боинг 737 MAX 7	2039	2
19	2025	12	Аэробус A350-1000	1967	3
20	2025	12	Боинг 787-9 Dreamli...	1494	4
21	2025	12	Эмбраэр E170	1123	5
22	2025	12	Аэробус A330-900neo	947	6

Всего строк: 32 Запрос завершен 00:00:00.189

Рисунок 10. Результат. Задание 5